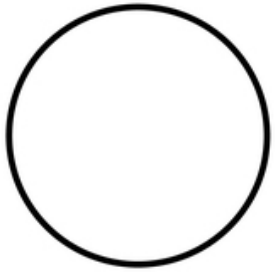
U(2)-Invariant Ricci Solitons on Four-Manifolds

William Hadden

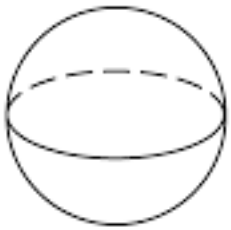
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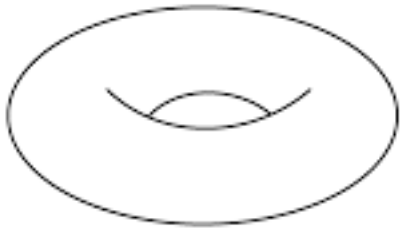
Topology

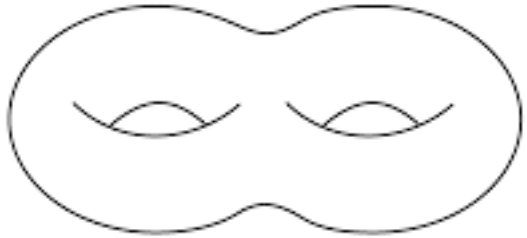
In 1 dimension:



In 2 dimensions:







In 1904, Poincaré conjectured that every three-dimensional topological manifold which is closed, connected and simply connected is homeomorphic to the three-dimensional sphere.

Geometry

In 3-dimensions, we have an equivalence of categories, **TopMan** $\simeq$ **SmoothMan**. Hence we are not concerned with changes in category in this case.

Then geometry enters the picture via the following theorem.

Theorem

Every smooth manifold, M admits a (non-canonical) Riemannian metric g.

For instance, we can distinguish between:

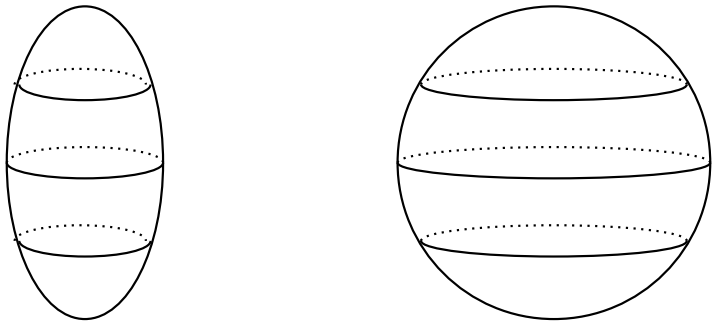


Figure: The 2-dimensional ellipsoid and S^2.

The *key* points are:

The Ricci Flow

The Ricci flow, evolves a Riemannian metric, g_0 by generating a sequence of time dependent metrics $\{g_t\}_{t\in I}$ via

$$\frac{\partial g_t}{\partial t} = -2\mathrm{Ric}(g_t)$$

where $\mathrm{Ric}(g_t)$ is the Ricci curvature of g_t. Heuristically, the Ricci flow spreads curvature out in space. For instance:

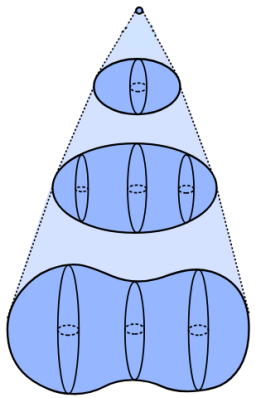


Figure: Ricci flow on a 3-manifold (Bamler, 2022).

- Compact 3-manifolds with positive Ricci curvature converge to metrics of constant sectional curvature (Hamilton, 1982)

Ricci Solitons

Ricci solitons are self similar solutions to the Ricci flow that evolve only by scalings and diffeomorphisms that satisfy

$$\mathrm{Ric}(g_0) + \frac{1}{2}\mathcal{L}_X(g_0) = \lambda g_0$$

These evolve as below.

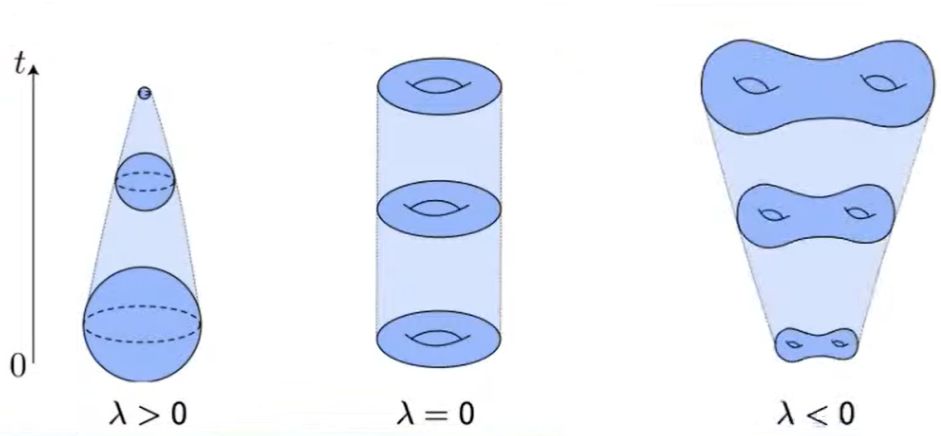


Figure: 3-dimensional Ricci solitons for different values of λ (Bamler, 2022).

Singularities and Poincaré Conjecture

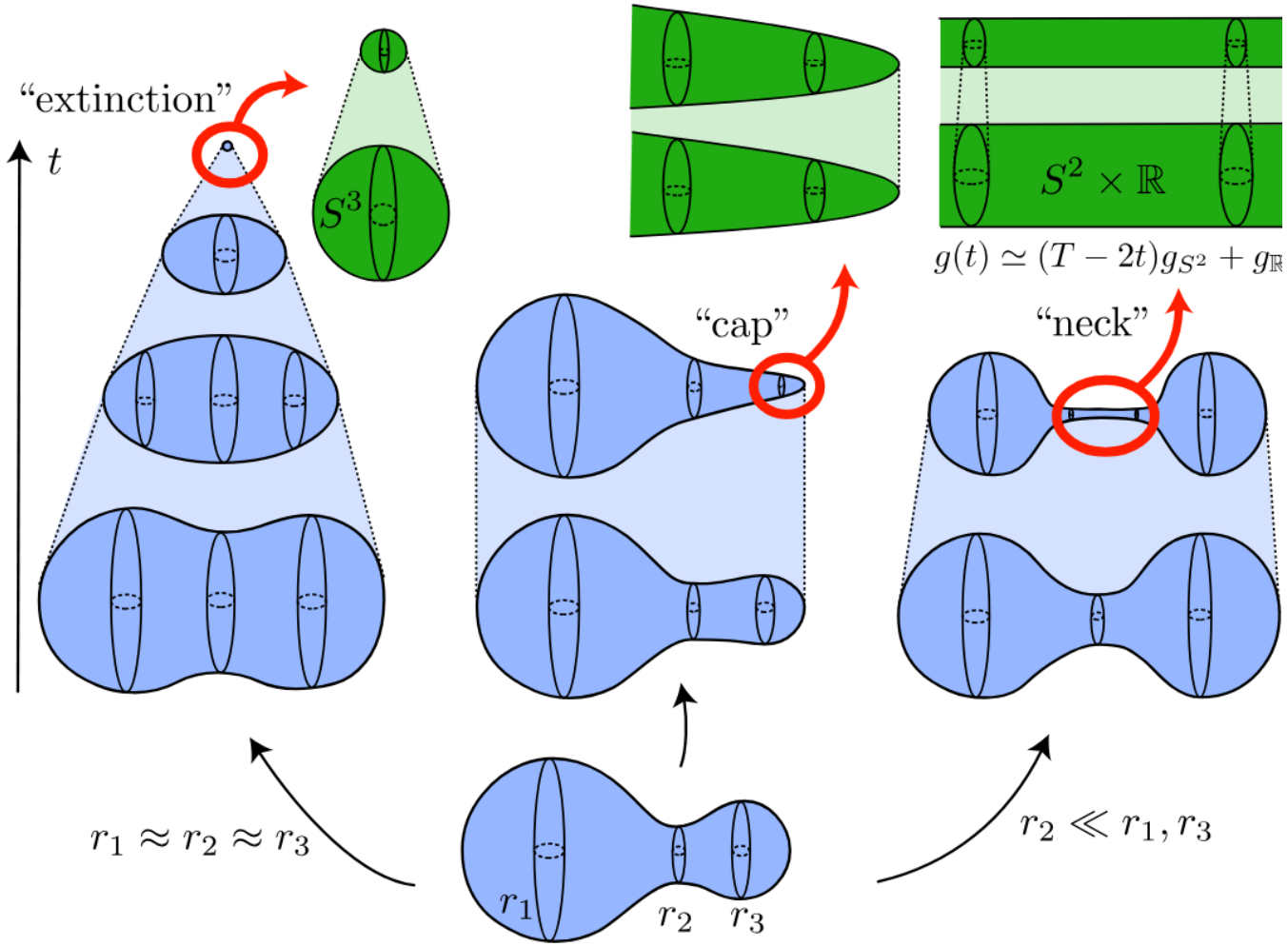
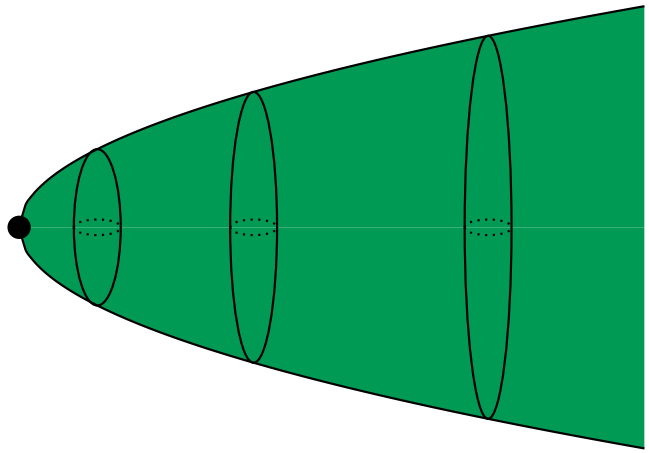


Figure: Singularity formation in the Ricci flow on a 3-sphere (Bamler, 2022)

Our Manifold

Our manifold of interest, M is diffeomorphic to the completion of $\mathbb{R}_{>0} \times S^3/\mathbb{Z}_k$ by adding a S^2 at the origin as pictured below.



We equip M with an action of $SU(2)$. The key idea is that $\{0\} \times S^2$ is a *singular orbit* of the left action of $SU(2)$ on M. Consider the map,

$$\pi : SU(2)/\mathbb{Z}_k \to SU(2)/U(1)$$
$$g\mathbb{Z}_k \mapsto gU(1)$$

such that $\pi^{-1}(gU(1)) \simeq U(1)/\mathbb{Z}_k$. That is,

Geometric Structure

We can equip the (principal component) $M_0 := (0, \infty) \times S^3/\mathbb{Z}_k$ with a metric of the form:

$$g_{M_0} = dr^2 + f_1(r)^2 g_S + f_2(r)^2 g_V$$

where g_S, g_V are given by the fibration, such that g_{M_0} is invariant under the action of $U(2)$ on M, denoted $\phi_h : U(2) \times M \to M$. That is,

$$\phi_h^* g = g$$

for all $h \in U(2)$.

Ricci Soliton Equations

The Ricci soliton equations become

$$u'' = \frac{f_1'}{f_1} + 2\frac{f_2'}{f_2} \tag{1}$$

$$\frac{d}{dr}\frac{f_1'}{f_1} = -(-u' + u'')\frac{f_1'}{f_1} + \frac{f_1^2}{2f_2^2} \tag{2}$$

$$\frac{d}{dr}\frac{f_2'}{f_2} = -(-u' + u'')\frac{f_2'}{f_2} + \frac{1}{f_2^2} - \frac{f_1^2}{2f_2^4}. \tag{3}$$

where $f_i : (0,\infty) \to (0,\infty)$, $f_i : r \mapsto f_i(r)$. To smoothly extend across the singular orbit at $r = 0$, we require:

$$f_1(0) = 0,\ f_1'(0) = 1,\ \text{and}\ f_2(0) = \overline{f_2} > 0,\ f_2'(0) = 0$$
$$u(0) = 0,\ u'(0) = 0,\ u''(0) = -\alpha \leq 0.$$

Two key points:

- The system (1)-(3) is degenerate at 0.
- Solutions to (1)-(3) depend continuously on α.

Numerical Solutions to the Soliton Equations

As the solutions depend (continuously) on α, we can numerically integrate the soliton equations to find:

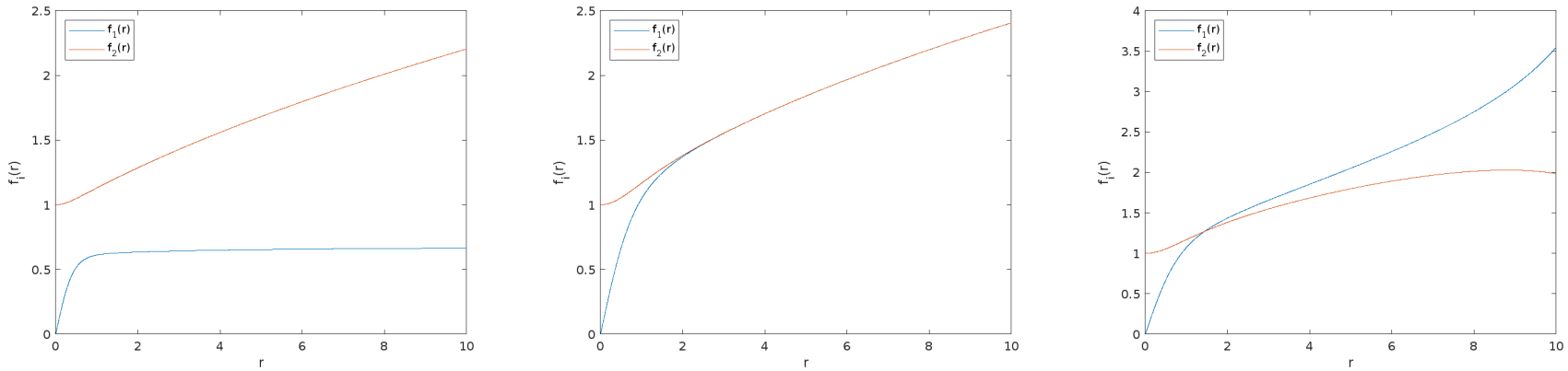


Figure: Collapsed, non-collapsed and incomplete solitons on $\mathbb{R}_{>0} \times SU(2)/\mathbb{Z}_3$.

A key scale invariant quantity of soliton equations is:

$$Q := \frac{f_1^2}{f_2^2}.$$

(Appleton, 2017) showed that:

- if $\lim_{r\to\infty} Q = 1$, then the solution is complete and

A. Appleton's Conjecture

As solutions to the soliton equations depend continuously on α, A. Appleton conjectured that for all $k \geq 3$ there exists a unique $\alpha^* \in \mathbb{R}$ such that the corresponding solitons are:

1. incomplete for all $\alpha > \alpha^*$,
2. complete and non-collapsed when $\alpha = \alpha^*$,
3. complete and collapsed when $\alpha < \alpha^*$.

A Change of Variables

(Buttsworth, 2021) examined the change of variables,

$$X_1 = \frac{1}{-u' + tr(L)} \frac{f_1'}{f_1}, \quad Y_1 = \frac{1}{-u' + tr(L)} \frac{f_1}{f_2^2}$$
$$X_2 = \frac{1}{-u' + tr(L)} \frac{f_2'}{f_2}, \quad Y_2 = \frac{1}{-u' + tr(L)} \frac{1}{f_1}$$

on the time re-parametrisation defined by $\frac{d}{ds} = \frac{1}{-u'+tr(L)} \frac{d}{dr}$. The two key features for our purposes are:

- The non-collapsed soliton satisfies $(X_1, X_2, Y_1, Y_2) \to (0, 0, 0, 0)$.
- $(0, 0, 0, 0)$ is a critical point in the variables (X_1, X_2, Y_1, Y_2).

The Center Manifold at the Origin

The analysis of the dynamics in some small neighbourhood of $(0, 0, 0, 0)$ can be described by Y_1, Y_2.

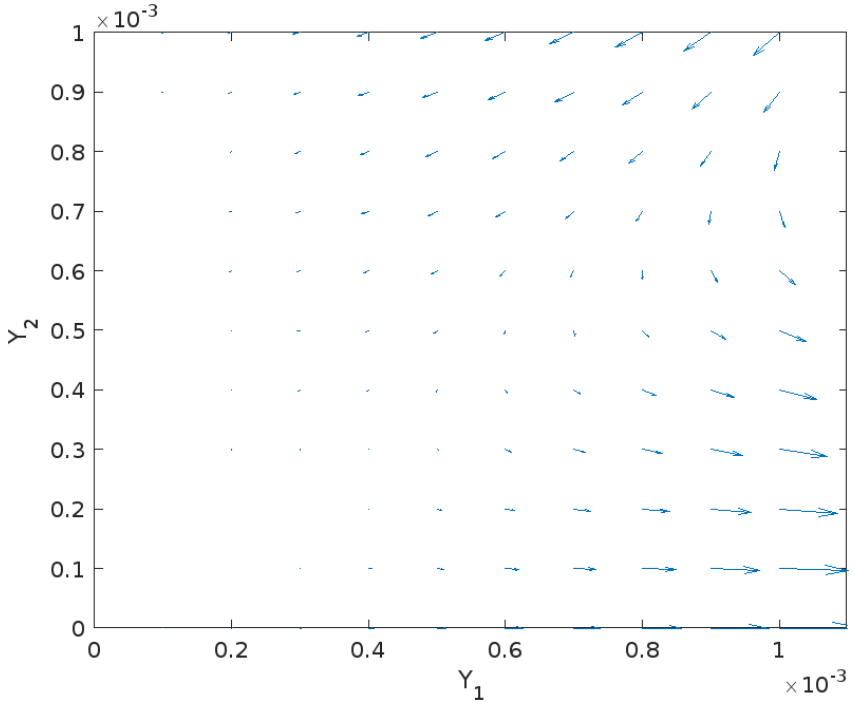


Figure: Vector field plot correct to second order about $(0, 0, 0, 0)$.

In particular,

- if $Y_1 > Y_2$, the solution is incomplete and blows up in finite time.

Numerical Calculation of the Center Manifold

Let $\mathcal{M}$ denote this neighbourhood of $(0,0,0,0)$. Then we can describe $\mathcal{M}$ via

$$\mathcal{M} = \{\varphi(z_c) \mid z_c \in \text{span}\{e_3, e_4\},\ |z_c| < \delta\}$$

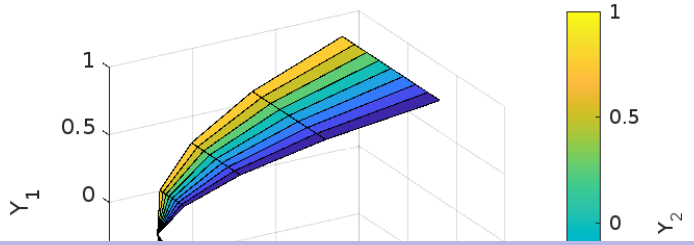
where

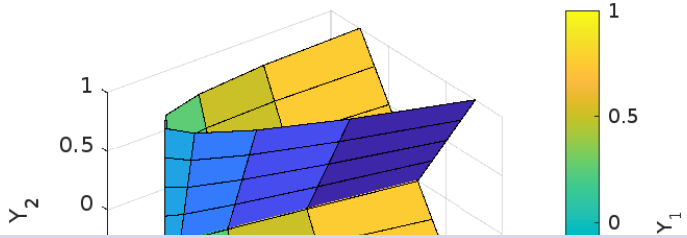
- $\varphi : z_c \mapsto y^{x_c}(0)$,
- y^{x_c} is the unique fixed point of the map $\Gamma : V^c \times \mathcal{Y} \to \mathcal{Y}$,
- Y is a Banach space

Γ is defined as:

$$\Gamma(x_c, y)(t) = e^{At}x_c + \int_0^t e^{A(t-\tau)}\pi_c g(y(\tau))d\tau + \int_{-\infty}^t e^{A(t-\tau)}\pi_s g(y(\tau))d\tau.$$

where $A \in M_4(\mathbb{R})$. Then $\mathcal{M} \subset \mathbb{R}^4$ is given by:





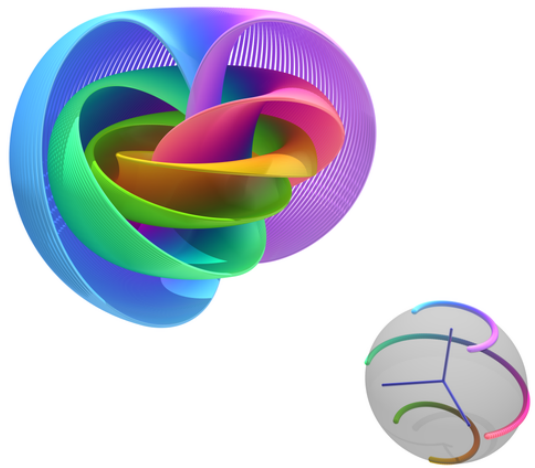
The Hopf Fibration

M is foliated by $S^3/\mathbb{Z}_k$ spheres with an S^2 cap. The resultant geometric structure is due to the Hopf fibration.

(Hopf, 1931) considered

$$S^1 \hookrightarrow S^3 \xrightarrow{\pi} S^2$$

where (the fibre) S^1 is embedded in (the total space) S^3 and π is a projection that projects S^3 onto (the base space) S^2. Then S^3 locally looks like $S^1 \times S^2$. The below image shows points on S^2 and their corresponding fibres in the same colour.



Our Manifold (in more detail)

We considered a smooth manifold of the form:

$$M = SU(2) \times_{U(1)} \mathbb{R}^2$$

with the equivalence relation $[g, x] \simeq [gh^{-1}, hx]$ where $g \in SU(2)$, $h \in U(1)$ and $x \in \mathbb{R}^2$.

Consider the map,

$$\pi : SU(2)/\mathbb{Z}_k \to SU(2)/U(1)$$
$$g\mathbb{Z}_k \mapsto gU(1)$$

such that $\pi^{-1}(gU(1)) \simeq U(1)/\mathbb{Z}_k$. That is,

$$U(1)/\mathbb{Z}_k \hookrightarrow SU(2)/\mathbb{Z}_k \xrightarrow{\pi} SU(2)/U(1).^2$$

Then our manifold of interest, M is the associated bundle to the ~~fibration π.~~

[2] Besse, A. L. (1987). Einstein manifolds. Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge · Band 10. Springer-Verlag, Berlin.